

Banking Credit Risk & Explainable Loan Decisioning Platform

Model governance report for a deterministic synthetic retail unsecured lending portfolio.

APPLICATIONS

48,000

Synthetic portfolio

OOT ROC AUC

0.738

2025 temporal holdout

DISPOSITION

REVIEW

No real lending use

Version 1.0.0 | July 2026 | Prepared as a professional portfolio project

SYNTHETIC DATA - DECISION SUPPORT ONLY - NO CLAIM OF REGULATORY COMPLIANCE

Executive summary

OOT ROC AUC

0.738Gate ≥ 0.700

Gini

0.475

Rank ordering

Approval recommendation

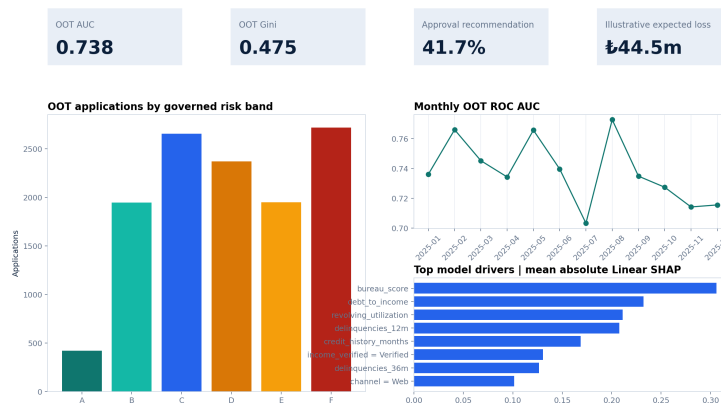
41.7%PD $< 10\%$

Management conclusion

The platform demonstrates a complete model-risk pattern: reproducible synthetic data, temporal development and validation, an interpretable champion, calibrated PDs, explicit policy thresholds, exact additive Linear SHAP, fairness diagnostics and monthly monitoring. The age-band allocation gap is intentionally surfaced as a governance blocker.

- Out-of-time ROC AUC is 0.738; calibration slope is 0.981.
- Approve recommendations cover 41.7% of OOT applications; their observed default rate is 7.9%.
- Linear SHAP reconstructs model log-odds with maximum error $2.66e-15$.
- The maximum demographic parity difference is 36.7%; governance review is required.
- The current disposition is portfolio-complete but not approved for real lending use.

Aurelia Bank | Credit Risk Decision-Support Control Tower



Synthetic portfolio • Recommendations are decision support only • Human review and jurisdiction-specific legal validation required

Figure 1. Executive control tower generated from deterministic OOT evidence.

Sources: internal artifacts; governance context [S1], [S2], [S4].

Purpose, scope and safety boundary

The implementation is an educational and portfolio-grade decision-support system for a fictional institution. It cannot issue a final approval, decline, price, limit or legally sufficient adverse-action notice. The API returns a recommendation that must be interpreted inside an authorized human workflow.

In scope	Mandatory oversight	Explicitly out of scope
12-month PD and score A-F risk bands Policy recommendations Illustrative expected loss Reason codes and monitoring	Credit officer review Independent model validation Fairness investigation Policy exception process Legal and disclosure review	Automated final decisions Real customer data Pricing or limits Production adverse action Claimed legal or accounting compliance

Human authority boundary

- The model estimates risk; policy maps the estimate to a recommendation.
- The review workflow can request evidence, override a recommendation or decline to act.
- Protected attributes are absent from the model and online contract.
- A production implementation would require jurisdiction-specific legal analysis, validation and approval.

Current model disposition: not approved for real credit decisioning. Fairness and external-validity work remain open.

Reference context: EU AI Act [S3], EBA loan origination guidelines [S2], NIST AI RMF [S4].

Portfolio and temporal design



Data-generation controls

- Deterministic NumPy generator with a fixed seed and canonical SHA-256 evidence.
- Application dates span 2022-01-01 through 2025-12-31.
- Target is a synthetic 12-month default indicator generated from non-protected risk drivers.
- The committed reviewer sample contains 250 rows; full portfolios are compressed CSV files.
- Six critical data-quality and safety controls pass before model fitting.

Control	Actual	Expected	Status
Expected schema present	28	28	PASS
Application ID unique	48000	48000	PASS
Target binary	0,1	0,1	PASS
Model feature completeness	0	0	PASS
Synthetic-only safety flag	True	True	PASS
Production decisions disabled	False	False	PASS

Internal sources: config/project.yaml, src/credit_risk/data.py, artifacts/metrics/data_quality_checks.csv.

Model development and champion selection

Champion policy
Logistic regression is selected because it leads the validation comparison on ROC AUC and Brier while supporting exact additive explanations. Histogram gradient boosting and random forest remain challengers.

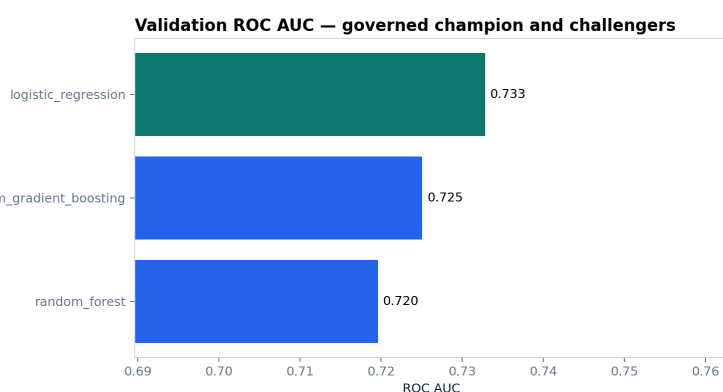


Figure 2. Validation ROC AUC.

Model	ROC AUC	Gini	Brier	Status
histogram gradient boosting	0.725	0.450	0.127	CHALLENGER
random forest	0.720	0.439	0.128	CHALLENGER
logistic regression	0.733	0.466	0.126	CHAMPION

Selection gates

- Temporal validation AUC ≥ 0.700 .
- Calibration slope within 0.75 to 1.25.
- Protected attributes excluded.
- Exact explanation identity test passes.
- Independent validation still required before any controlled pilot.

Internal sources: src/credit_risk/modeling.py and artifacts/metrics/model_comparison.csv.

Out-of-time performance

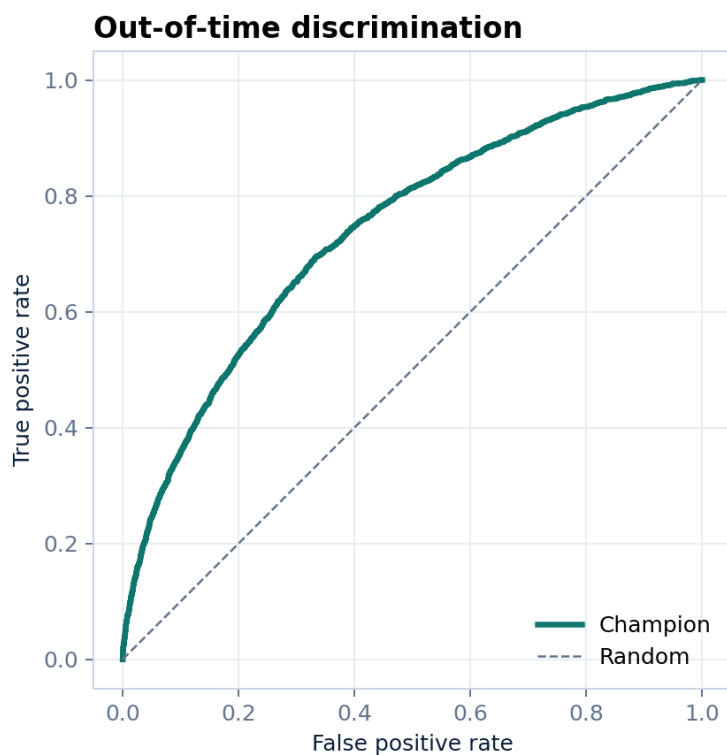


Figure 3. OOT ROC curve.

ROC AUC

0.738

Gate ≥ 0.700

Gini

0.475

$2 \times \text{AUC} - 1$

KS

0.361

Max separation

PR AUC

0.421

Default-focused

The champion crosses the predefined synthetic-project discrimination gate with a credible margin. The result is evidence for this generated portfolio only; it cannot be extrapolated to a real applicant population without representative data, redevelopment and validation.

Internal source: artifacts/metrics/sample_metrics.json. Metric implementation reference: scikit-learn [S7].

Calibration and risk deciles

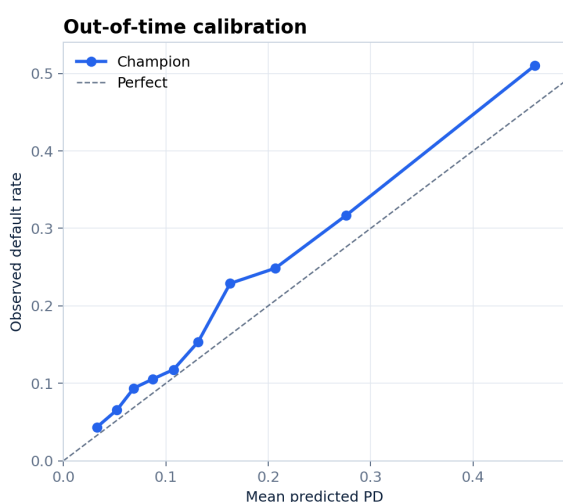


Figure 4. OOT calibration.

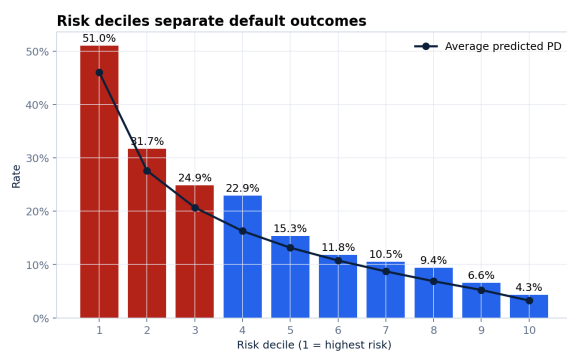


Figure 5. OOT risk deciles.

Calibration slope

0.981

Target 0.75 to 1.25

Brier

0.134

Lower is better

OOT default rate

18.8%

Observed

Interpretation

- Calibration evaluates whether probability levels match observed event rates.
- Risk deciles test ordering and default capture across the score distribution.
- Brier combines calibration and discrimination; it is not a calibration-only metric.
- Monthly monitoring repeats both performance and distribution checks.

Calibration reference: scikit-learn probability calibration [S7]. Internal decile table: artifacts/metrics/risk_deciles.csv.

Decision policy and score mapping

PD interval	Recommendation	Required action
PD < 10%	APPROVE_RECOMMENDATION	Human confirmation may still apply
10% <= PD < 22%	REFER	Mandatory authorized review
PD >= 22%	DECLINE_RECOMMENDATION	Mandatory review and disclosure workflow

Score and expected-loss conventions

- Score anchor is 650 points at 10% PD; 50 points represent a doubling of odds.
- Scores are clipped to 300 to 850 and bands A to F communicate continuous PD.
- Illustrative expected loss equals $PD \times 45\% \text{ LGD} \times \text{requested amount}$.
- The expected-loss bridge is not an IFRS 9 impairment model: it has no staging, lifetime PD, discounting or macroeconomic scenarios.
- The Excel workbench exposes policy thresholds as editable inputs with formula-driven outputs.

Policy ownership: model development owns estimation evidence; the credit policy committee owns thresholds; authorized officers own final decisions.

Risk bands and illustrative expected loss

Band	Applications	Average PD	Observed DR	Expected loss
A	421	2.4%	3.3%	TRY 0.21m
B	1,948	4.6%	5.8%	TRY 1.88m
C	2,657	7.9%	10.1%	TRY 4.58m
D	2,372	12.3%	14.2%	TRY 6.61m
E	1,949	18.1%	23.4%	TRY 7.80m
F	2,718	35.3%	39.8%	TRY 23.38m

OOT EL

TRY 44.5m

Illustrative

Approval rate

41.7%

PD < 10%

Approved DR

7.9%

Observed

The band table helps a committee understand portfolio mix, risk concentration and the consequences of changing the approve boundary. Scenario outputs remain illustrative because approval propensity, pricing response, prepayment and recoveries are not modeled.

Explainability and reason-code governance

Global Linear SHAP importance

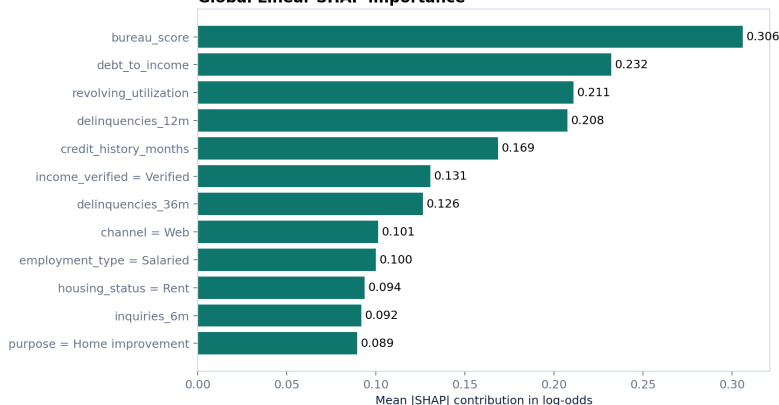


Figure 6. Global Linear SHAP importance.

Exact additive identity

For the standardized and one-hot encoded logistic model, the implementation computes $\phi_{ij} = \beta_j \times (x_j - E[x_j])$ in log-odds space. The expected log-odds plus all feature contributions reconstruct the model output to numerical precision.

Max additivity error

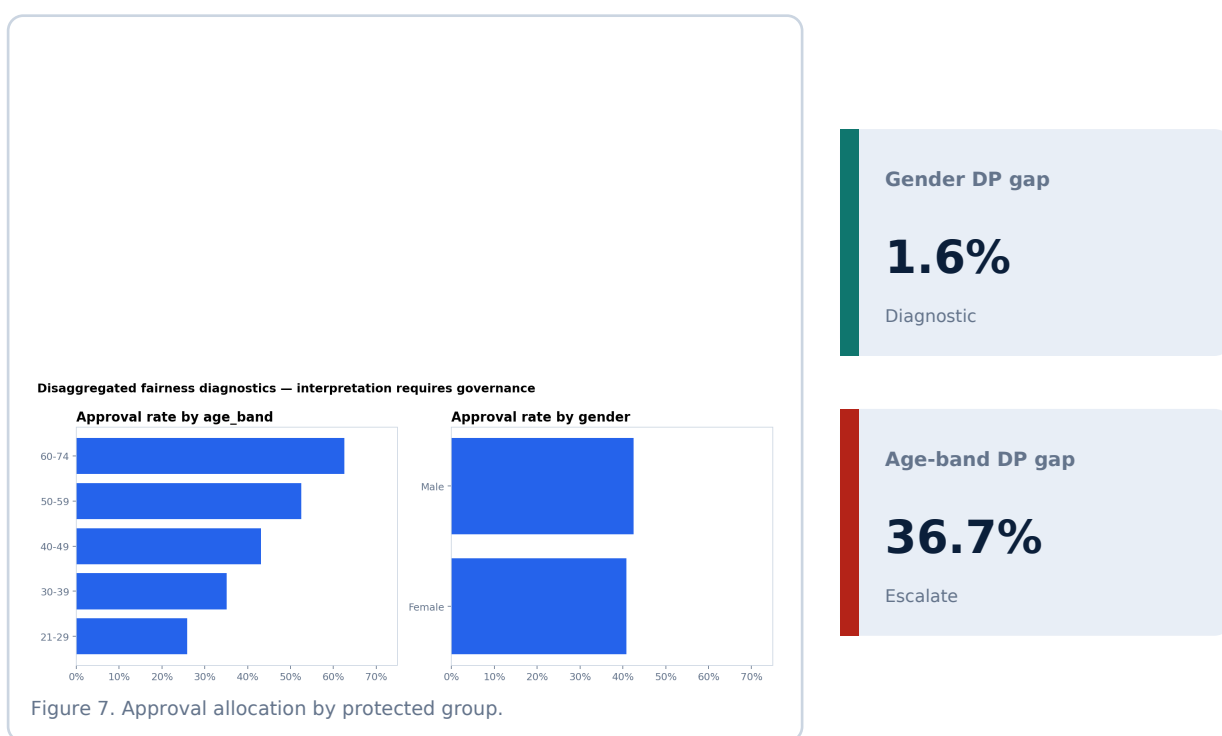
2.7e-15

Log-odds

Reason-code contract

- Return the top four risk-increasing contributions for reviewer attention.
- Keep protected attributes out of the model and explanation state.
- Describe contributions as model evidence, never causal findings.
- Do not treat model reason codes as a legally sufficient adverse-action notice.
- Preserve the full contribution table for validation and audit.

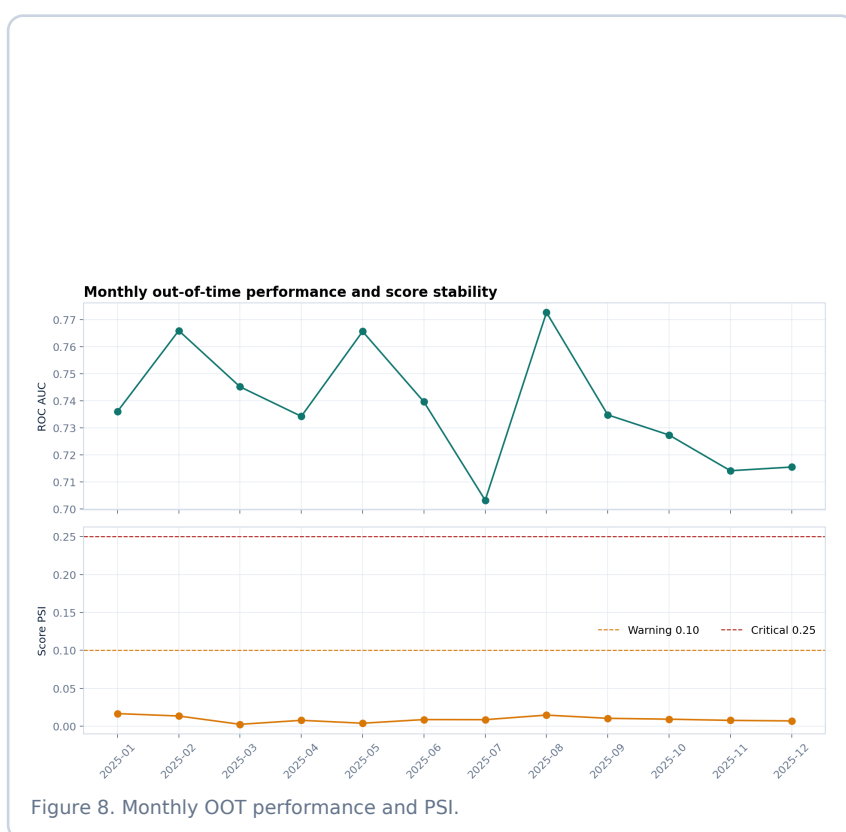
Fairness assessment and governance disposition



GOVERNANCE REVIEW REQUIRED. The age-band difference exceeds the project's internal 10 percentage-point triage signal. This threshold is not a legal fairness standard.

- Investigate credit-history and tenure pathways that may act differently across age groups.
- Review sample composition, legitimate risk factors, policy thresholds and measurement uncertainty.
- Perform intersectional and jurisdiction-specific analysis with legal and stakeholder input.
- Do not suppress the finding or call the model fair because protected attributes were excluded.

Model monitoring and incident response



Max score PSI

0.017

Warning 0.10

Months

12

2025 OOT

Monitoring controls

- Monthly ROC AUC, Brier, observed default rate and approval recommendation rate.
- Score PSI plus bureau-score and DTI population-stability diagnostics.
- Green, amber and red triage with explicit investigation and rollback guidance.
- API throughput, latency, errors and availability exposed to Prometheus.
- Model incidents require preserved inputs, output, version, policy state and human action.

Monitoring references: Evidently drift docs [S9], Prometheus instrumentation [S10]. Internal runbook: docs/operations/monitoring-runbook.md.

Architecture, API and security controls

Layer	Implementation	Primary control
Analytics	Python package and deterministic pipeline	Temporal split, metrics, SHAP identity
Decision support	FastAPI /api/v1/score	Strict schema; protected fields absent
Data	PostgreSQL schemas and views	Governance schema separation
BI	Power BI project and Excel workbench	Formula lineage and source audit
Operations	Prometheus and Grafana	Availability, latency and errors
Runtime	Docker and Kubernetes	Non-root, read-only, capabilities dropped
Delivery	GitHub Actions	Tests, coverage, CodeQL, pip-audit, Trivy

API response boundary

The scoring endpoint returns request ID, model version, 12-month PD, credit score, risk band, recommendation, human-review flag, illustrative expected loss and four reason codes. It does not return or accept protected attributes and does not execute an external action.

- API-key option and extra-field rejection.
- Health, readiness and Prometheus endpoints.
- No credentials or production endpoints committed.
- Hardened container and restrictive Kubernetes network policy.

Internal sources: Dockerfile, docker-compose.yml, infra/kubernetes/, src/credit_risk/api.py and service.py.

Reference mapping - not a compliance claim

Primary sources inform the control design. Applicability and conformity require qualified legal, accounting, risk and validation review.

ID	Primary source	Control influence	URL
S1	Basel Committee - Principles for management of credit risk (2025)	Granting, monitoring, controls	bis.org/bcbs/publ/d595.pdf
S2	EBA - Guidelines on loan origination and monitoring	Creditworthiness governance	eba.europa.eu/.../EBA GL 2020 06
S3	Regulation (EU) 2024/1689 - AI Act	High-risk context and oversight	eur-lex.europa.eu/eli/reg/2024/1689/oj
S4	NIST AI RMF 1.0	Govern, Map, Measure, Manage	nist.gov/itl/ai-risk-management-framework
S5	IFRS 9 Financial Instruments	Expected-credit-loss terminology	ifrs.org/issued-standards/.../ifrs-9
S6	SHAP documentation	Additive explanation method	shap.readthedocs.io/en/latest/
S7	scikit-learn calibration documentation	Calibration curves and Brier context	scikit-learn.org/stable/modules/calibration.html
S8	Fairlearn assessment documentation	Disaggregated fairness metrics	fairlearn.org/main/user_guide/assessment/
S9	Evidently data drift documentation	Distribution shift concepts	docs.evidentlyai.com/metrics/preset_data_drift
S10	Prometheus instrumentation guidance	Operational metrics design	prometheus.io/docs/practices/instrumentation/

These references are crosswalk inputs. The project does not certify compliance with Basel standards, EBA guidelines, the EU AI Act, IFRS 9 or any consumer-credit law.

Validation evidence, limitations and residual risk

Control	Result	Disposition
Data quality	6 / 6 PASS	Proceed to model evidence
Temporal OOT AUC	0.738	Synthetic gate passes
Calibration slope	0.981	Within 0.75 to 1.25
SHAP identity	2.7e-15	Passes numerical tolerance
Monthly score PSI	max 0.017	Below 0.10 warning
Fairness	max DP 36.7%	Governance review required
Workbook controls	8 / 8 PASS	Formula and source checks pass
Slide layout	20 / 20 PASS	No overflow detected

Material limitations

- All data are synthetic; no representativeness or production validity is established.
- Target generation and model form are simplified and share an intentionally controlled structure.
- No reject inference, bureau data licensing, fraud controls or affordability policy is implemented.
- Expected loss is illustrative and not an IFRS 9 engine.
- Fairness metrics are descriptive and do not resolve legal, causal or normative questions.
- No independent validator or production change-approval committee has reviewed the model.

Internal QA evidence: artifacts/manifest.json, reports/workbook/, reports/presentation/ and tests/.

Türkçe yönetici özeti ve sonraki adımlar

Proje, 48.000 sentetik kredi başvurusu üzerinde 12 aylık temerrüt olasılığı, kredi skoru, A-F risk segmentasyonu, SHAP neden kodları, politika önerileri ve model izleme katmanlarını tek bir yönetim mimarisinde birleştirir. 2025 dönemindeki zaman dışı testte ROC AUC 0.738, Gini 0.475 ve kalibrasyon eğimi 0.981 olarak ölçülmüştür.

Temel sonuç

- Model ve politika ayrılmış, nihai karar yetkisi insan incelemesinde tutulmuştur.
- Korunan nitelikler modelden çıkarılmış ve sadece ayrı adalet denetiminde kullanılmıştır.
- SHAP katkıları log-odds düzeyinde matematiksel olarak doğrulanmıştır.
- Yaş grupları arasındaki belirgin tahsis farkı, gerçek kullanım öncesi zorunlu yönetim bulgusudur.
- Proje portföy seviyesi açısından tamamdır; gerçek kredi kararları için onaylı değildir.

Kontrollü pilot için yol haritası

Aşama	Çalışma	Çıkış kriteri
1	Bağımsız doğrulama	Metrikleri yeniden üret, varsayımları zorla
2	Adalet incelemesi	Yaş grubu farkının nedenlerini ve vekil etkileri araştır
3	Hukuk ve politika	Yetki, bildirim, itiraz ve override akışını onayla
4	Shadow çalışma	Aksiyon üretmeden izleme ve olay tatbikatı yap
5	Komite kararı	Sınırlı kapsam, rollback ve periyodik doğrulama

Son durum: Profesyonel portföy paketi tamamlandı. Gerçek müşteri verisi veya otomatik kredi kararı için kullanılamaz.